

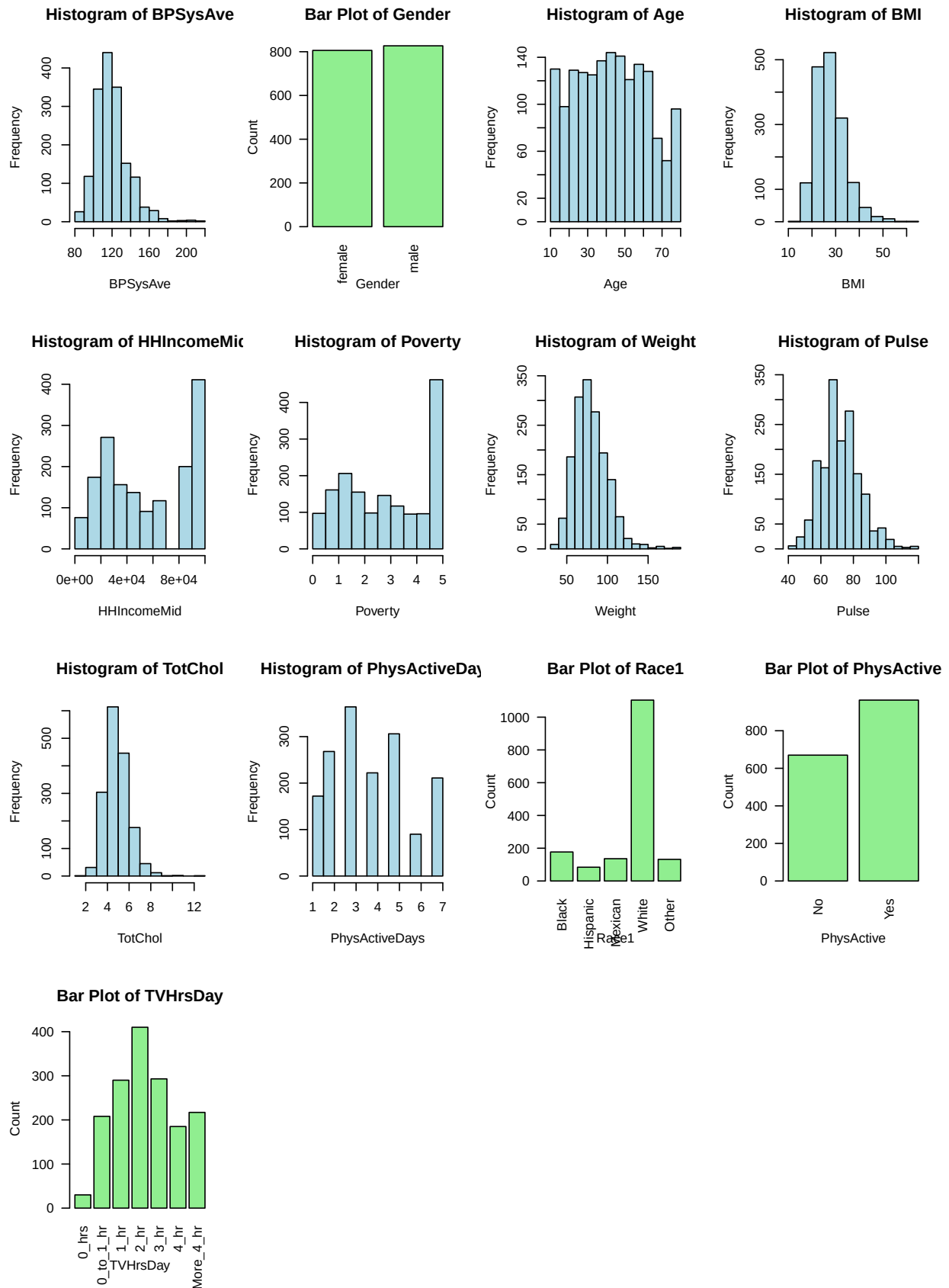
STA302 Report

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Importing Data from NHANES package

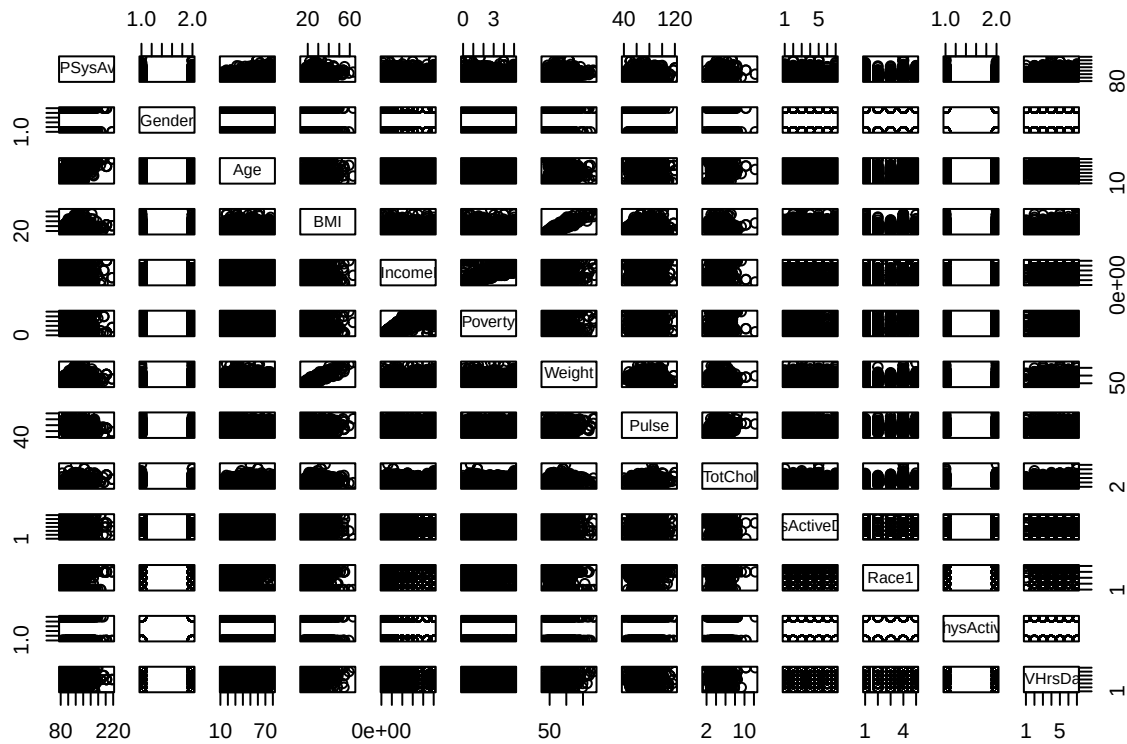
```
library(NHANES)
data(NHANES)
# select columns we want from NHANES
data <- subset(as.data.frame(NHANES),
               select = c(BPSysAve, Gender, Age, BMI,
                          HHIncomeMid, Poverty, Weight, Pulse, TotChol, PhysActiveDays, Race1, PhysActiveDays))
data = na.omit(data)
```

Plot the data to see distributions:



Checking correlation between predictors:

```
plot(data)
```



Summary of BPSysAve, our response variable

```
summary(data$BPSysAve)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      81.0   109.0   118.0   120.2   129.0   217.0
```

Graphing interaction terms:

```
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2) + 0.1)

# Make a scatterplot of blood pressure and age, coloured by gender
plot(data$Age, data$BPSysAve,
     main = "Interaction of Age and Gender on Blood Pressure",
     xlab = "Age",
     ylab = "Systolic Blood Pressure",
```

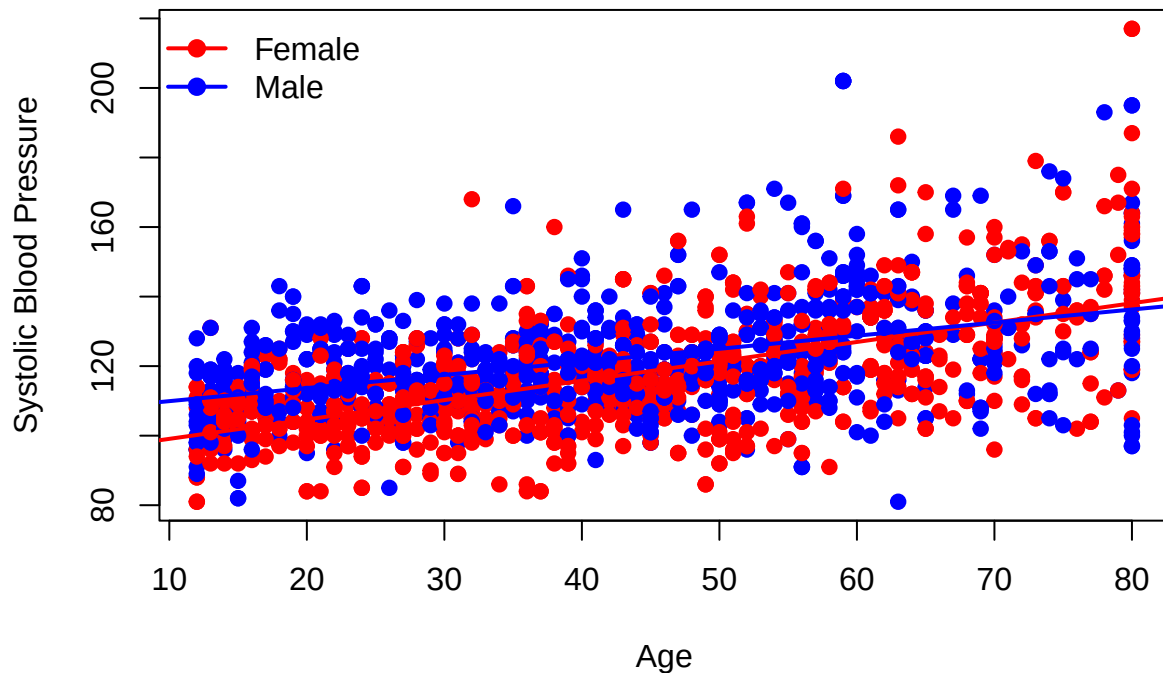
```

pch = 19,
col = ifelse(data$Gender == "male", "blue", "red"))
# Add female line of best fit
female_fit <- lm(BPSysAve ~ Age, data = subset(data, Gender == "female"))
abline(female_fit, col = "red", lwd = 2)
# Add male line of best fit
male_fit <- lm(BPSysAve ~ Age, data = subset(data, Gender == "male"))
abline(male_fit, col = "blue", lwd = 2)

legend("topleft",
      legend = c("Female", "Male"),
      col = c("red", "blue"),
      pch = 19, lwd = 2, bty = "n")

```

Interaction of Age and Gender on Blood Pressure



```

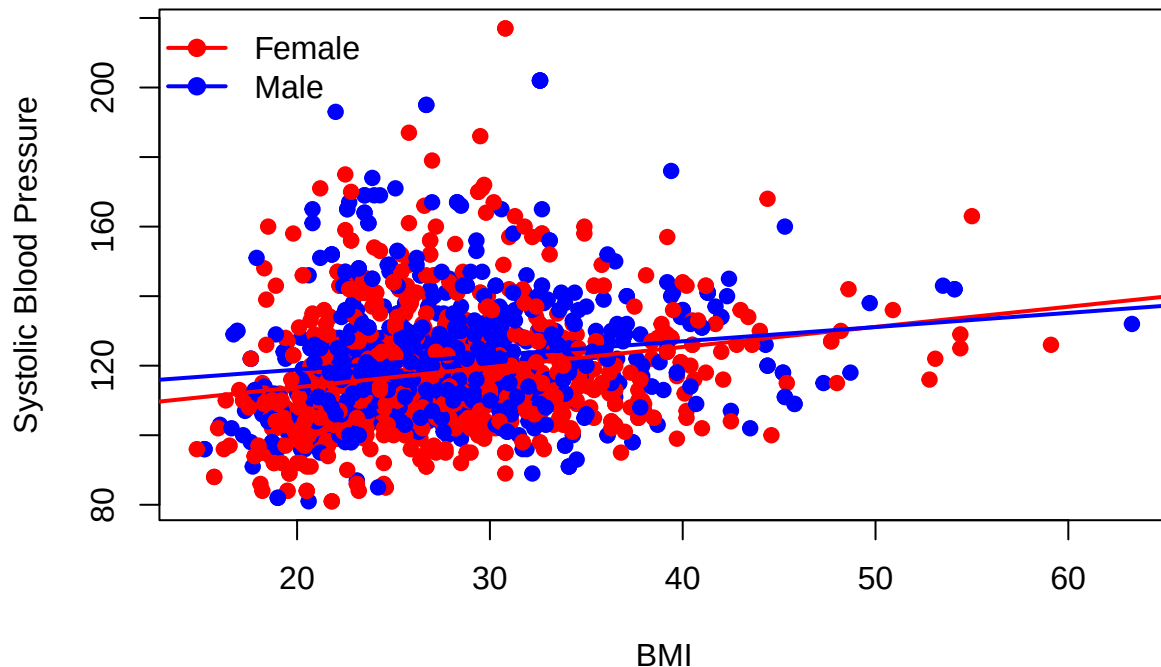
# Make a scatterplot of blood pressure and BMI
plot(data$BMI, data$BPSysAve,
      main = "Interaction of BMI and Gender on Blood Pressure",
      xlab = "BMI",
      ylab = "Systolic Blood Pressure",
      pch = 19,
      col = ifelse(data$Gender == "male", "blue", "red"))
# Add female line of best fit
female_fit_bmi <- lm(BPSysAve ~ BMI, data = subset(data, Gender == "female"))
abline(female_fit_bmi, col = "red", lwd = 2)
# Add male line of best fit

```

```
male_fit_bmi <- lm(BPSysAve ~ BMI, data = subset(data, Gender == "male"))
abline(male_fit_bmi, col = "blue", lwd = 2)

legend("topleft",
      legend = c("Female", "Male"),
      col = c("red", "blue"),
      pch = 19, lwd = 2, bty = "n")
```

Interaction of BMI and Gender on Blood Pressure

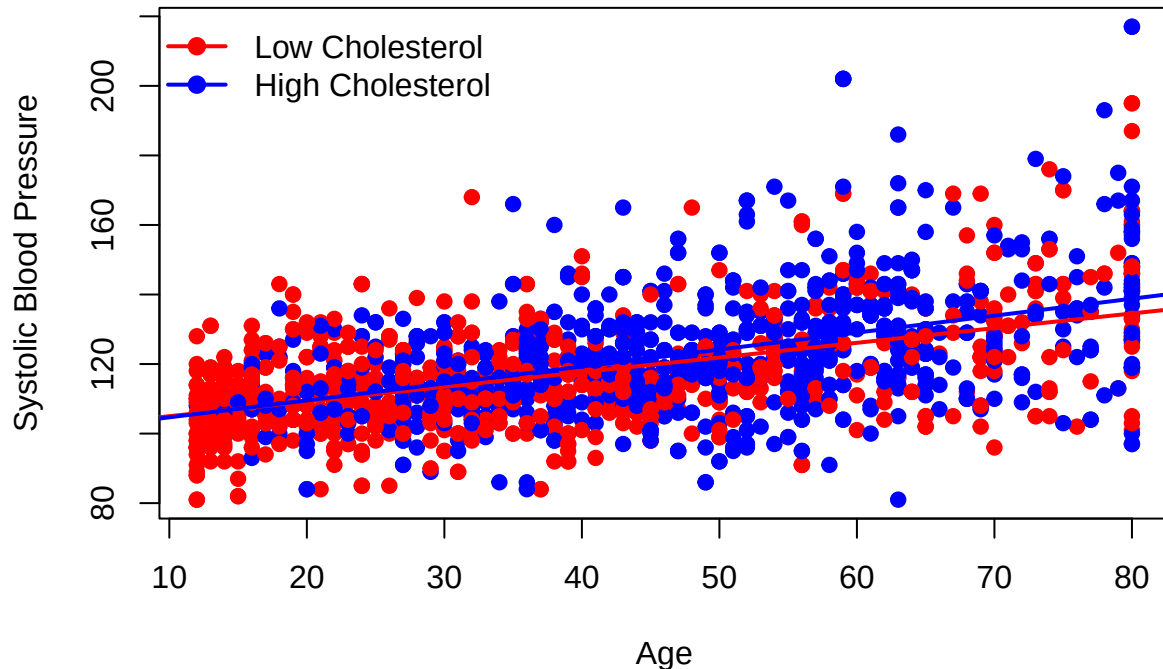


```
# Split TotChol into a high and low category using the median, we will use this to visualize different
median_chol <- median(data$TotChol)
```

```
# Make a scatterplot of blood pressure and cholesterol
plot(data$Age, data$BPSysAve,
      main = "Interaction of Age and Total Cholesterol on Blood Pressure",
      xlab = "Age",
      ylab = "Systolic Blood Pressure",
      pch = 19,
      col = ifelse(data$TotChol < median_chol, "red", "blue")) #colour based on group
# Add low cholesterol line of best fit
low_chol_fit <- lm(BPSysAve ~ Age, data = subset(data, TotChol < median_chol))
abline(low_chol_fit, col = "red", lwd = 2)
# Add high cholesterol line of best fit
high_chol_fit <- lm(BPSysAve ~ Age, data = subset(data, TotChol >= median_chol))
abline(high_chol_fit, col = "blue", lwd = 2)
```

```
legend("topleft",
      legend = c("Low Cholesterol", "High Cholesterol"),
      col = c("red", "blue"),
      pch = 19, lwd = 2, bty = "n")
```

Interaction of Age and Total Cholesterol on Blood Pressure



```
par(mfrow = c(1, 1))
```

Fitting our preliminary model

```
# Fit a model with our predictors, as well as some interesting interaction terms
model = lm(BPSysAve ~ . + Gender:Age + Gender:BMI + Age:TotChol, data=data)
summary(model)
```

```
##
## Call:
## lm(formula = BPSysAve ~ . + Gender:Age + Gender:BMI + Age:TotChol,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.884  -8.593  -0.827   7.356  75.763
##
```

```

## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    8.860e+01  6.292e+00  14.083 < 2e-16 ***
## Gendermale     1.511e+01  3.424e+00   4.412 1.09e-05 ***
## Age           2.703e-01  9.934e-02   2.721 0.006574 **
## BMI            1.938e-01  1.557e-01   1.245 0.213490
## HHIncomeMid    1.123e-04  2.328e-05   4.825 1.53e-06 ***
## Poverty       -3.336e+00  4.701e-01  -7.096 1.91e-12 ***
## Weight         4.019e-02  5.077e-02   0.792 0.428760
## Pulse          8.174e-02  3.156e-02   2.590 0.009682 **
## TotChol       -1.812e+00  9.757e-01  -1.857 0.063429 .
## PhysActiveDays -2.732e-01  1.924e-01  -1.420 0.155913
## Race1Hispanic  -6.355e+00  1.925e+00  -3.302 0.000980 ***
## Race1Mexican   -4.583e+00  1.669e+00  -2.747 0.006089 **
## Race1White     -4.419e+00  1.199e+00  -3.687 0.000234 ***
## Race1Other     -3.676e+00  1.689e+00  -2.176 0.029681 *
## PhysActiveYes   7.830e-01  7.810e-01   1.003 0.316187
## TVHrsDay0_to_1_hr 4.234e+00  2.798e+00   1.513 0.130445
## TVHrsDay1_hr    4.575e+00  2.745e+00   1.667 0.095717 .
## TVHrsDay2_hr    5.845e+00  2.702e+00   2.163 0.030698 *
## TVHrsDay3_hr    5.344e+00  2.746e+00   1.947 0.051754 .
## TVHrsDay4_hr    5.138e+00  2.818e+00   1.824 0.068412 .
## TVHrsDayMore_4_hr 4.446e+00  2.784e+00   1.597 0.110494
## Gendermale:Age  -1.561e-01  3.872e-02  -4.033 5.76e-05 ***
## Gendermale:BMI  -1.254e-01  1.162e-01  -1.079 0.280906
## Age:TotChol     6.605e-02  1.952e-02   3.384 0.000731 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.19 on 1609 degrees of freedom
## Multiple R-squared:  0.3465, Adjusted R-squared:  0.3372
## F-statistic: 37.1 on 23 and 1609 DF, p-value: < 2.2e-16

```

Residual Plots:

